

Answer Key

Quiz 2 - Retake

Quiz Code: **482768**

Quiz Information

Total Questions: 20

Time Limit: 15 minutes

Quiz Period: Mar 12, 2026 13:40 - Mar 12, 2026 14:40

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All Questions with Correct Answers

Question 1

1. In a Hard-Margin SVM, what is the primary constraint for all training points (x_i, y_i) ?

A $y_i(w^T x_i + b) \geq 1$

✓ Correct Answer

B $y_i(w^T x_i + b) \geq 0$

C $w^T x_i + b = 0$

D $y_i(w^T x_i + b) \leq 1$

Question 2

2. To maximize the margin $M = \frac{2}{\|w\|}$, what objective function do we minimize?

A $\frac{1}{2} \|w\|$

B $\frac{1}{2} \|w\|$

✓ Correct Answer

C $\sum \xi_i$

D $w^T x + b$

Question 3

3. Which points in the dataset actually determine the position of the decision boundary?

A All points in the dataset

B Only the outliers

C The Support Vectors

✓ Correct Answer

D Only the points at the origin

Question 4

4. What does a "Hard Margin" SVM do if the data is not linearly separable?

A It finds the best possible fit

B It ignores the outliers

C It fails to find a valid solution

✓ Correct Answer

D It automatically applies a kernel

Question 5

5. The distance from a point x to the hyperplane $w^T x + b = 0$ is given by:

A $\frac{\|w^T x + b\|}{\|w\|}$

✓ Correct Answer

B $w^T x + b$

C $\frac{1}{2} \|w\|$

D $\|w\|$

Question 6

6. In Soft-Margin SVM, what is the purpose of the slack variable ξ_i ?

A To increase the weight of features

B To allow points to be inside the margin or misclassified

✓ Correct Answer

C To project data into higher dimensions

D To calculate the bias term

Question 7

7. What is the objective function for a Soft-Margin SVM?

A $\frac{1}{2} \|w\|$

B $\sum \xi_i$

C $\frac{1}{2} \|w\|^2 + C \sum \xi$

✓ Correct Answer

D $w^T x + b + C$

Question 8

8. In the Soft-Margin parameter C , what happens as $C \rightarrow \infty$?

A The margin becomes wider

B It behaves like a Hard-Margin SVM

✓ Correct Answer

C All points become support vectors

D The model ignores all constraints

Question 9

9. What is the "Hinge Loss" formula used in SVMs?

A $\min(0, 1 - y_i f(x_i))$

B $\max(0, 1 - y_i f(x_i))$

✓ Correct Answer

C $(y_i - f(x_i))^2$

D $\log(1 + e^{-y_i f(x_i)})$

Question 10

10. Geometrically, if $\xi_i > 1$, where is the data point located?

A On the correct side of the margin

B Exactly on the decision boundary

C On the wrong side of the decision boundary

✓ Correct Answer

D Exactly on the margin

Question 11

11. The "Kernel Trick" allows us to compute the dot product in high-dimensional space using:

- A Explicit feature mapping $\phi(x)$
- B A function $K(x_i, x_j)$ in the original space** ✓ Correct Answer
- C Only the bias term
- D Iterative gradient descent

Question 12

12. What is the formula for the Gaussian (RBF) Kernel?

- A $x_i^T x_j$
- B $(x_i^T x_j + 1)^d$
- C $\exp(-\gamma \|x_i - x_j\|^2)$** ✓ Correct Answer
- D $\tanh(\kappa x_i^T x_j + c)$

Question 13

13. In the dual form, the decision function $f(x)$ depends only on:

- A The inverse of the covariance matrix
- B Dot products between input vectors** ✓ Correct Answer
- C The magnitude of the weights
- D The total number of features

Question 14

14. If you use a Polynomial Kernel with $d = 2$ on 2D data (x_1, x_2) , the feature space becomes:

- A 2-dimensional
- B 3-dimensional
- C 6-dimensional** ✓ Correct Answer
- D Infinite-dimensional

Question 15

15. Why is the Dual form preferred over the Primal form for the Kernel trick?

A It has fewer variables

B It involves only dot products of x_i, x_j ✓ Correct Answer

C It eliminates the need for C

D It doesn't require support vectors

Question 16

16. For a linear kernel, the weight vector w can be recovered from the dual as:

A $w = \sum \alpha_i y_i x_i$ ✓ Correct Answer

B $w = \sum \alpha_i x_i$

C $w = \sum y_i x_i$

D $w = \max(\alpha_i)$

Question 17

17. What is the dimensionality of the feature space for a Linear Kernel?

A Same as input space

✓ Correct Answer

B Infinite

C One less than input space

D Zero

Question 18

18. Which of these is NOT a standard SVM kernel?

A Sigmoid

B Laplacian

C Logistic

✓ Correct Answer

D Linear

Question 19

19. How does the Kernel Trick affect computational complexity regarding the feature space dimension D ?

- A** Complexity grows linearly with D
- B** Complexity grows exponentially with D
- C** Complexity is independent of D ✓ Correct Answer
- D** Complexity is zero

Question 20

20. Which of the following is true about the SVM margin?

- A** It is the area where data is most dense
- B** It is the "no-man's land" between classes ✓ Correct Answer
- C** It is only defined for 2D data
- D** It is maximized by increasing $\|w\|$

